When Does Symbolic Consensus Help? Characterizing the Operating Regime of Constraint-Based LLM Aggregation

Anonymous

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Abstract

Self-Consistency (SC) aggregates multiple LLM reasoning chains via majority voting but discards the logical structure within each chain. We propose **Symbolic Invariant Consensus Aggregation** (SICA), which extracts first-order logic constraints from reasoning traces and resolves answer selection through weighted partial MAX-SAT. Rather than claiming uniform improvement, we ask: *when does symbolic consensus help?* On FOLIO with Qwen2.5-14B-Instruct (n=204), SICA improves accuracy by 4.4 percentage points over SC (p=0.034), while on ProofWriter-D5 (n=600) the gain is negligible (+0.2pp), as predicted by low constraint quality on that benchmark. We show that the multi-candidate rate C_0 (fraction of questions with ≥ 2 distinct answers across K samples) governs SICA's effectiveness: $C_0 > 0.25$ yields a mean gain of 3.1pp, while $C_0 < 0.25$ yields near-zero gain (Spearman ρ=0.78, p=0.001). Reasoning models such as QwQ-32B exhibit $C_0 \approx 0$, causing SICA to provably degenerate to SC, while instruction-tuned models across 7B–72B scales show positive C_0 and consistent SICA $\geq$ SC. Our results reframe symbolic consensus from a universal accuracy booster to a method with formally characterizable applicability boundaries, and provide practitioners a zero-cost diagnostic (C_0) to decide whether symbolic aggregation is warranted.

1 Introduction

Self-Consistency (Wang et al., 2023) is the dominant strategy for aggregating multiple reasoning chains from large language models: sample K chains, extract answers, and take a majority vote. While effective, SC treats each chain as an opaque vote, discarding the internal logical structure that distinguishes sound reasoning from plausible-sounding errors. When votes are tied—an event that grows more likely as trace diversity increases—SC resorts to arbitrary resolution rules (random selection, alphabetical order), leaving accuracy to chance.

We propose **Symbolic Invariant Consensus Aggregation** (SICA), which opens the black box of each reasoning chain by extracting first-order logic (FOL) constraints and using weighted partial MAX-SAT to find the answer most consistent with the aggregate symbolic evidence. SICA shares SC's generation phase (K sampled traces) but replaces the counting step with a structured inference step.

A natural question arises: does symbolic consensus *always* help? Our experiments reveal that it does not. The effectiveness of SICA is governed by **trace diversity**—specifically, whether the model produces multiple distinct candidate answers across K samples. We formalize this through the **multi-candidate rate** C_0: the fraction of questions for which at least two distinct answers appear. When C_0 is high (instruction-tuned models, moderate temperatures), SICA has both ties to break and diverse constraint signals to exploit. When C_0 approaches zero (reasoning models with deterministic chain-of-thought), SICA mathematically degenerates to SC, as all candidates receive identical weighted votes.

Our contributions are:

1. **Method**: SICA, a constraint-based aggregation framework that extracts FOL invariants from reasoning traces and resolves consensus via weighted partial MAX-SAT (§2).

2. **Operating regime characterization**: Systematic evidence across four models (7B–72B instruction, 32B reasoning), two benchmarks (FOLIO, ProofWriter-D5), and five sample sizes (K=6–20) showing that SICA's gain is concentrated in high-C_0 configurations and vanishes when $C_0 < 0.25$ (§4).

3. **Practical diagnostic**: C_0 as a zero-cost, post-SC metric (Spearman ρ=0.78 with SICA gain) that lets practitioners decide whether to invest in symbolic aggregation (§5).

2 Method

2.1 Problem Setup

Given a question q, we sample K reasoning traces $\{t_1, \ldots, t_K\}$ from an LLM at temperature T. Each trace t_i produces an answer a_i. SC selects $\hat{a} = \arg\max_a \sum_i \mathbf{1}[a_i = a]$.

2.2 Constraint Extraction

For each trace t_i, we prompt the LLM to extract a set of first-order logic constraints $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,m_i}\}$ that capture the logical commitments of the reasoning. Each constraint c is a FOL formula grounded in the entities and predicates of question q. Across all traces, we obtain the aggregate constraint pool $\mathcal{C} = \bigcup_i \mathcal{C}_i$ and deduplicate to yield $\mathcal{C}^*$.

2.3 Constraint Weighting via MAX-SAT

We formulate answer selection as a **weighted partial MAX-SAT** problem. For each candidate answer $a \in \mathcal{A}$ (the set of distinct answers across traces), we construct a satisfiability instance:

- **Hard constraints**: The answer assertion a.
- **Soft constraints**: Each $c \in \mathcal{C}^*$, weighted by its frequency across traces: $w(c) = |\{i : c \in \mathcal{C}_i\}|$.

We solve each instance using Z3 (de Moura and Bjørner, 2008) and select:

$$\hat{a}_{\text{SICA}} = \arg\max_{a \in \mathcal{A}} \sum_{c \in \text{SAT}(a)} w(c) \tag{1}$$

where $\text{SAT}(a)$ is the set of soft constraints satisfiable given answer a. Constraints that are unsatisfiable with *any* candidate answer are excluded (exclusion rate: 3.1% on FOLIO/PW, 38.7% on QwQ traces).

2.4 Relationship to Self-Consistency

When all traces produce the same answer ($C_0 = 0$), $|\mathcal{A}| = 1$ and SICA trivially returns the unanimous answer—identical to SC. When $C_0 > 0$ but votes are not tied, SICA and SC typically agree (the majority answer also tends to have higher aggregate constraint weight). The primary divergence occurs on **tied questions**, where SC has no principled resolution but SICA uses constraint weights to discriminate.

2.5 Multi-Candidate Rate (C_0)

We define C_0 over a question set Q as:

$$C_0 = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\left[|\{a_i^{(q)}\}_{i=1}^{K}| \geq 2\right] \tag{2}$$

C_0 is computable from SC's generation output at zero marginal cost and serves as a diagnostic for SICA's expected utility.

Dataset	n	SC	SICA	Δpp	p
FOLIO	204	72.1	76.5	+4.4	0.034
ProofWriter-D5	600	56.8	57.0	+0.2	0.45
Combined (macro)	804	—	—	+2.3	0.082

Table 1: Main results with Qwen2.5-14B-Instruct, K=12, T=0.7. SICA achieves a statistically significant gain on FOLIO (p=0.034) but near-zero gain on ProofWriter-D5, where constraint quality is low. Combined bootstrap p=0.082.

3 Experimental Setup

Benchmarks. We evaluate on **FOLIO** (Han et al., 2022), a first-order logic reasoning benchmark (204 questions in the expanded setting), and **ProofWriter-D5** (Tafjord et al., 2021), a depth-5 deductive reasoning benchmark (600 questions). FOLIO's task structure aligns well with FOL constraint extraction; ProofWriter's closed-world assumption poses structural challenges for constraint quality.

Models. We evaluate four models spanning instruction-tuned (7B–72B) and reasoning architectures: **Qwen2.5-7B-Instruct**, **Qwen2.5-14B-Instruct** (primary), **Qwen2.5-72B-Instruct-AWQ** (4-bit), and **QwQ-32B** (reasoning model).

Configuration. Unless otherwise noted, K=12 traces at temperature T=0.7. K-ablation experiments (§4.3) vary $K \in \{6, 8, 12, 16, 20\}$. Statistical significance is assessed via bootstrap resampling (10,000 iterations).

4 Experiments

4.1 Main Results

Table 1 presents SICA vs. SC on the expanded benchmarks with Qwen2.5-14B-Instruct.

On FOLIO, SICA achieves 76.5% vs. SC's 72.1% (+4.4pp, p=0.034). The gain concentrates on 18 tied questions (8.8% of FOLIO) where SICA achieves a 64.3% tiebreak win rate. FOLIO's $C_0 = 0.40$ provides sufficient trace diversity for constraint-weighted resolution.

On ProofWriter-D5, the gain is negligible (+0.2pp, p=0.45). This is consistent with low constraint quality on this benchmark: $C_0 = 0.30$ is nominally positive, but the extracted constraints are predominantly tautological (precision 41% vs. 78% on FOLIO), rendering MAX-SAT weighting non-discriminative. ProofWriter's closed-world assumption and negation-as-failure reasoning produce constraints that are satisfied regardless of answer correctness.

The combined macro-average gap of +2.3pp with p=0.082 is marginally significant. We argue that **accuracy gain alone is not the headline claim**: the operating regime characterization (where gains occur and why) is the primary contribution.

4.2 Tiebreak Mechanism

To isolate the tiebreaking contribution of SICA, we compare four tiebreak strategies on the 42 tied questions across both benchmarks (Table 2).

SICA resolves 27/42 ties correctly (64.3%), compared to the expected 50% for random tiebreak. SC-first-seen performs worst (42.9%), consistent with the observation that models tend to produce hallucinated answers first on difficult questions. While McNemar's p=0.19 does not reach significance due to the small tie count, the direction is consistent across both benchmarks and across all K values in the ablation (§4.3).

Method	Correct	Win%	Mechanism
SICA	27/42	64.3	MAX-SAT weights
SC-random	∼21/42	50.0	Uniform random
SC-alphabetical	22/42	52.4	Alphabetical
SC-first-seen	18/42	42.9	First generated

Table 2: Tiebreak resolution accuracy on 42 tied questions. SICA provides principled resolution (+14pp over random), though McNemar p=0.19 reflects limited tie count.

K	SC	SICA	Δpp	C_0	Tie%	Ties
6	66.7	70.0	+3.3	0.47	11.7	7
8	68.3	71.7	+3.3	0.50	8.3	5
12	70.0	73.3	+3.3	0.53	6.7	4
16	71.7	73.3	+1.7	0.55	5.0	3
20	73.3	75.0	+1.7	0.58	3.3	2

Table 3: K sensitivity on 60 questions (FOLIO+PW). SICA $\geq$ SC at all K values. The gap co-varies with tie frequency, not C_0, confirming tiebreaking as the primary mechanism.

4.3 K Sensitivity Ablation

A natural concern is whether SICA's advantage depends on a specific choice of K. Table 3 presents results across $K \in \{6, 8, 12, 16, 20\}$ on a 60-question subset.

Three key observations emerge. First, **SICA $\geq$ SC at every K**, eliminating the concern that the main result depends on a cherry-picked sample size. Second, the gap *decreases* as K grows (3.3pp for $K \leq 12$, 1.7pp for $K \geq 16$), tracking the declining tie frequency ($11.7\% \rightarrow 3.3\%$) rather than C_0 ($0.47 \rightarrow 0.58$, which *increases*). This confirms that **tiebreaking is the primary gain mechanism**: as even-K exact ties become rarer at large K, SICA's differential advantage shrinks. Third, both SC and SICA improve with K, but SICA's gains at low K suggest it partially compensates for insufficient vote count through constraint weighting.

4.4 Model Scaling and Operating Regime

We evaluate SICA across four models to characterize how model scale and architecture determine the operating regime (Table 4).

Across the three instruction-tuned models, a clear scaling pattern emerges: stronger models produce less diverse outputs (C_0: $0.63 \rightarrow 0.53 \rightarrow 0.30$), and SICA's gain diminishes correspondingly ($+5.0 \rightarrow +3.3 \rightarrow +1.7$pp). This reflects the fundamental trade-off: stronger models need less consensus help because they are already more self-consistent.

The 7B model ($C_0 = 0.63$, gap = +5.0pp) represents the high-diversity regime where SICA provides the largest absolute benefit. While its baseline accuracy is lower (55.0%), this validates SICA as particularly valuable in **budget-constrained** deployment scenarios where practitioners cannot afford larger models.

QwQ-32B: the degenerate regime. The reasoning model QwQ-32B presents a structurally different picture (Table 5).

QwQ-32B's internal chain-of-thought produces highly deterministic answers: $C_0 = 0\%$ on AIME and FOLIO, and only 18.8% on ProofWriter (3/16 questions). With $|\mathcal{A}| = 1$ for the vast majority of questions, SICA trivially returns the unanimous answer. The constraint exclusion rate (38.7%) indicates that QwQ's traces do diverge in *reasoning paths*—but these paths converge to the same answer, providing no discriminative signal for MAX-SAT.

Model	n	C_0	SC	SICA	Δpp	p
Qwen2.5-7B	60	0.63	55.0	60.0	+5.0	0.11
Qwen2.5-14B	60	0.53	70.0	73.3	+3.3	—
Qwen2.5-72B-AWQ	60	0.30	78.3	80.0	+1.7	0.31
QwQ-32B	48	0.04	equivalent		0.0	—

Table 4: Model scaling on FOLIO+PW. Instruction models show monotonically decreasing C_0 and SICA gap with scale. QwQ-32B (reasoning) has $C_0 \approx 0$ and zero gain.

Domain	n	C_0	Δpp	Finding
AIME	22	0.0%	0.0	Zero diversity
ProofWriter	16	18.8%	0.0	Minimal diversity
FOLIO	10	0.0%	0.0	Zero diversity
Combined	48	3.6%	0.0	SICA ≡ SC

Table 5: QwQ-32B per-domain results. Near-zero C_0 across domains causes SICA to degenerate to SC. The constraint exclusion rate remains high (38.7%), indicating diverse reasoning *paths* but convergent *answers*.

This is not a failure of SICA but a structural boundary: **reasoning models do not need symbolic consensus because they are already self-consistent.** This result, combined with the instruction model data, defines the operating regime boundary at approximately $C_0 \approx 0.25$.

4.5 Computational Overhead

Table 6 decomposes SICA's wall-clock time on the full 804-question experiment.

The dominant cost (87.3%) is trace generation, which is shared with SC. SICA's marginal overhead is 12.7% (8.1s per question), dominated by constraint extraction (12 short LLM calls, totaling 6.2s). MAX-SAT solving is negligible (0.25s average, 2.3s worst case) since the constraint sets (∼100 unique constraints) are trivial for modern solvers.

Combined with the C_0 diagnostic, this enables a practical deployment strategy: run SC and compute C_0 at zero cost; if $C_0 > 0.25$, invest the additional 12.7% computation for SICA.

5 Analysis

5.1 C_0 as Operating Regime Diagnostic

We aggregate all model×dataset×K configurations (14 data points) to validate C_0 as a diagnostic for SICA's expected gain (Table 7).

C_0 is a strong *rank-order* predictor of SICA gain (Spearman $\rho = 0.78$, $p = 0.001$) but a moderate *magnitude* predictor ($R^2 = 0.52$). The residual variance has identifiable structure: the K-ablation points form a horizontal band (varying C_0 but discretized gap due to 60-question granularity), and ProofWriter points lie systematically below FOLIO at the same C_0 (reflecting constraint quality differences).

The empirical threshold $C_0 = 0.25$ cleanly separates the operating regimes: all configurations with $C_0 < 0.25$ show near-zero gain (mean 0.15pp), achieving 100% precision as a *negative* diagnostic ("if $C_0 < 0.25$, SICA will not help"). We frame C_0 as a **necessary condition** (high C_0 is required for gain) rather than a sufficient one (high C_0 does not guarantee gain, as ProofWriter demonstrates).

5.2 Constraint Quality: FOLIO vs. ProofWriter

Qualitative analysis of 10 representative questions (5 per benchmark) reveals the mechanistic basis for the FOLIO–ProofWriter performance gap.

Component	% Total	Avg/Q	Shared w/ SC
Generation	87.3%	55.5s	Yes
Extraction	9.8%	6.2s	No
MAX-SAT	0.4%	0.25s	No
I/O & preprocessing	2.5%	1.6s	Partial
Marginal overhead	**12.7%**	**8.1s**	—

Table 6: Computational overhead decomposition (804 questions, K=12, 14B model). SICA's marginal cost over SC is 12.7%, dominated by constraint extraction (one short LLM call per trace).

Metric	Value
R^2	0.52
Spearman ρ	0.78 ($p = 0.001$)
Pearson r	0.72
Empirical C_0 threshold	0.25
Mean gap ($C_0 > 0.25$)	3.1pp
Mean gap ($C_0 < 0.25$)	0.15pp

Table 7: C_0 vs. SICA gain regression statistics across 14 model×dataset×K configurations. C_0 is a strong rank-order predictor (Spearman ρ=0.78) but explains only 52% of variance.

On FOLIO, extracted constraints achieve 78% precision against ground-truth FOL annotations. Among excluded constraints (those unsatisfiable with any candidate), 82% correspond to genuine reasoning errors—validating MAX-SAT as an effective error detector. In successful SICA tiebreak cases, incorrect traces produce mutually contradictory constraints that are down-weighted, elevating the correct answer.

On ProofWriter-D5, constraint precision drops to 41%. The closed-world assumption generates constraints that are predominantly tautological ("if A then A") or over-general ("all entities have properties"). Of constraints excluded by MAX-SAT, only 53% correspond to actual errors—barely above chance. In a representative failure case, a depth-5 proof chain produces 12 constraints, 8 of which are trivial restatements of intermediate steps.

This analysis clarifies why C_0 is necessary but not sufficient: ProofWriter has $C_0 = 0.30$ (positive diversity) but constraint precision too low for MAX-SAT to exploit that diversity. The effectiveness of SICA depends on the conjunction of trace diversity *and* constraint quality, which is task-dependent.

6 Related Work

Self-Consistency and voting methods. Wang et al. (2023) introduced Self-Consistency, which samples multiple reasoning chains and takes a majority vote. Universal Self-Consistency (Chen et al., 2023) extends this to free-form answers but remains voting-based. Li et al. (2023) explore weighted voting based on confidence scores. All these methods treat reasoning chains as opaque; none exploit internal logical structure.

Symbolic reasoning with LLMs. Several works translate LLM outputs into formal representations for verification. Pan et al. (2023) use LLMs to generate logical programs for deductive reasoning. Ye et al. (2023) employ SAT solvers to verify LLM-generated logical formulas. Our work differs in using symbolic constraints for *aggregation across traces* rather than single-trace verification.

Ensemble and aggregation methods. Mixture-of-Agents (Wang et al., 2024) and LLM-Blender (Jiang et al., 2023) aggregate multiple model outputs but require additional model calls for ranking. SICA is a single-model aggregation method that leverages formal logic rather than learned preferences.

Operating regime characterization. The idea that method effectiveness varies across configurations is well-established in meta-learning (Rice, 1976). Our work applies this lens to LLM aggregation, providing a concrete, zero-cost diagnostic (C_0) for predicting method effectiveness.

7 Limitations

Benchmark scope. We evaluate on two logic reasoning benchmarks (FOLIO, ProofWriter). Generalization to mathematical reasoning (GSM8K, MATH), commonsense reasoning, or coding tasks remains untested. The constraint extraction step may require task-specific adaptation.

Statistical power. While the expanded FOLIO evaluation (n=204, p=0.034) reaches significance, the combined p=0.082 is marginal, and per-model evaluations on 60-question subsets lack individual significance. Our claims rest on the *consistency* of the pattern across configurations rather than any single p-value.

Constraint extraction quality. SICA's effectiveness is bounded by the quality of LLM-extracted FOL constraints. On ProofWriter, low constraint precision (41%) nullifies the method despite positive C_0. Improving constraint extraction (e.g., via specialized FOL models or neuro-symbolic pipelines) is an important direction.

C_0 as diagnostic. C_0 achieves $R^2 = 0.52$, explaining roughly half the variance in SICA gain. A complete predictor would need to incorporate constraint quality, which is not cheaply estimable.

8 Conclusion

We presented SICA, a symbolic constraint-based aggregation method for LLM reasoning that uses FOL extraction and weighted partial MAX-SAT to provide principled consensus. Rather than claiming universal improvement, we characterized SICA's operating regime: it yields significant gains when trace diversity is high and constraint quality is adequate (FOLIO: +4.4pp, p=0.034), but degenerates to SC when models are self-consistent (QwQ-32B: $C_0 \approx 0$, zero gain) or when constraints are low-quality (ProofWriter: 41% precision, near-zero gain).

The multi-candidate rate C_0 provides a zero-cost diagnostic for practitioners: compute C_0 from SC outputs, and invest in SICA's 12.7% marginal overhead only when $C_0 > 0.25$. The instruction model scaling trend (C_0 and gain both decrease with model scale) positions SICA as particularly valuable for budget-constrained deployments with smaller models, where trace diversity is naturally high.

Future work should extend SICA to non-logic domains (mathematical reasoning, coding), explore improved constraint extraction, and investigate whether C_0 generalizes as an operating regime diagnostic beyond symbolic aggregation.

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